

Curriculum Vitae

WEI-CHE HSU

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EDUCATION

Arizona State University (ASU)

Doctor of Philosophy in Industrial Engineering

- Advisor: Dr. Zihan Zhang

Aug 2026 – Present

Tempe, Arizona

National Yang Ming Chiao Tung University (NYCU)

Master of Science in Industrial Engineering and Management

- Advisor: Dr. Wen-Chih Chen

Sep 2021 – Aug 2023

Hsinchu, Taiwan

National Tsing Hua University (NTHU)

Bachelor of Science in Industrial Engineering and Engineering Management

- Minor: Power Mechanical Engineering
- Certification: Data Science

Sep 2017 – Jun 2021

Hsinchu, Taiwan

PROFESSIONAL EXPERIENCE

Micron Technology – Industrial Engineer

Full-time, Department of Operations Intelligence

Mar 2024 – Jun 2026

Taichung, Taiwan

- Improved transfer efficiency in the Automated Material Handling System (AMHS)
 - Analyzed cross-fab transfer causes and redundant commands, developed improvement strategies, and contributed to a 63.6% reduction in redundant moves and a 36% reduction in AMHS-related tool idle time
 - Automated AMHS data collection and processing in Python and built Tableau dashboards for operational performance monitoring
- Optimized tool-step assignment strategies to mitigate cross-fab congestion under high-loading conditions
 - Constructed a discrete-event simulation model for tool-step assignment evaluation, reducing simulation runtime by 98.6% (from approximately 36 hours to 30 minutes)
 - Designed a customized Genetic Algorithm (GA) for tool-step assignment optimization and path-release strategy search, reducing cross-fab congestion while satisfying wafer-throughput requirements

Micron Technology – Industrial Engineer

Intern, Department of Operations Intelligence

Jul 2022 – Aug 2022

Taichung, Taiwan

- Analyzed key drivers of cross-fab AMHS transfer inefficiencies and proposed actionable improvement strategies
 - Created an AMHS performance metric and Tableau dashboards, with the metric later adopted for routine fab-level performance monitoring

Vanguard International Semiconductor – Data Analytics Engineer

Intern, Department of Production Controlling

Jul 2021 – Aug 2021

Hsinchu, Taiwan

- Built a regression-based production output forecasting model in Python, reducing mean relative error (MRE) by approximately 67% and achieving the target KPI of MRE < 15%

RESEARCH EXPERIENCE

A Simulation-Based Opt. Study of Procurement Lot Sizing under Demand Uncertainty

M.S. Thesis (2023), Department of Industrial Engineering and Management, NYCU

- Formulated a stochastic procurement lot-sizing model to optimize ordering and inventory decisions under uncertain demand
- Designed a simulation-optimization framework combining Genetic Algorithm (GA) and Simultaneous Perturbation Stochastic Approximation (SPSA), improving solution quality by over 10% on large-scale instances relative to either method alone

Energy Efficiency Opt. of Variable-Freq. Strategy for AC Systems in Wafer Mfg. Plant

Academia-Industry Collab. (2020), Department of Industrial Engineering and Engineering Management, NTHU

- Developed a variable-frequency control strategy for air-conditioning systems in semiconductor manufacturing
- Integrated regression models with Genetic Algorithm (GA) and Particle Swarm Optimization (PSO) to optimize energy consumption, achieving approximately US\$70K in monthly cost savings

TEACHING EXPERIENCE

Teaching Assistant — Course: Data Structure (Class Size: 60 Students)	Feb 2023 – June 2023
<i>Department of Industrial Engineering and Management, NYCU</i>	<i>Hsinchu, Taiwan</i>
• Provided student support on algorithm design, data structures, and programming concepts • Designed homework and exam questions aligned with course learning objectives • Graded coding assignments for 60 students and developed an automated grading system using predefined input/output cases to ensure consistent and accurate evaluation	

HONORS & AWARDS

Fulton Scholar IV, Ira A. Fulton Schools of Engineering , Arizona State University	2026-27
Third Place, Summer Intern Project Competition , Vanguard International Semiconductor Corp.	2021
Industrial Engineering Undergraduate Project Award , National Tsing Hua University	2020

RELEVANT COURSEWORK

Optimization & Operations Research: Operations Research I & II; Advanced Operations Research (<i>simulation optimization and algorithm evaluation</i>); Linear Programming (<i>simplex methods, duality, and sensitivity analysis</i>); Integer Programming & Network Analysis (<i>integer formulations and network-flow models</i>); Stochastic Programming (<i>scenario-based optimization under uncertainty</i>)
Stochastic Modeling & Statistics: Probability Theory; Engineering Statistics; Stochastic Processes (<i>Markov chains and probabilistic system modeling</i>)
Data Analytics & Computing: Data Structures; Big Data Analytics & Data Mining (<i>data-mining methods and R-based analysis</i>); Smart Data Analytics (<i>analytical methods for engineering and healthcare applications</i>)
Mathematical Foundations: Linear Algebra; Discrete Mathematics; Numerical Analysis (<i>approximation methods, error analysis, and numerical stability</i>)
Additional Engineering Coursework: Engineering Mathematics I (<i>ordinary differential equations</i>); Manufacturing Processes; Applications of the Finite Element Method (<i>finite-element modeling and engineering analysis</i>)

PROGRAMMING & ANALYTICAL TOOLS

Programming Languages: C/C++, C#, Python, R, MATLAB, JavaScript
Database: Microsoft SQL Server
Visualization: Tableau, Microsoft Power BI